

ETE 403

VLSI Design

Reference Notes

References: Brackenbury — VLSI Design · Pucknell & Eshraghian — Basic VLSI Design, 3rd Ed. · Mead & Conway — Introduction to VLSI Systems

Chittagong University of Engineering and Technology

Class 05

Md. Hasibul Islam Jihad

Department of ETE · Level 4, Term 1

Course Instructor: **Arif Istiaq**, Assistant Professor

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Chapter 2 (Continued): MOS Devices and Basic Circuits

Enhancement-type NMOS is the standard device used for MOSFET analysis. Its majority carriers are electrons, which are real physical particles and straightforward to visualize, making current flow easy to follow.

PMOS uses holes instead of electrons. Since holes are the absence of an electron rather than a real particle, current directions and voltage polarities become reversed, making analysis less intuitive.

Depletion-type MOSFETs are also less convenient because a channel already exists at zero gate voltage. Operation means removing a channel rather than forming one, which reverses many of the usual conditions.

For these reasons, enhancement-type NMOS is taken as the reference device throughout this course.

2.16 NMOS Transistor Structure

An NMOS transistor is built on a silicon wafer that acts as the substrate or body of the device. The wafer is made of ultra-pure silicon and is doped with P-type impurities, forming a P-type substrate.

Since the device is an N-channel MOSFET, both the Source and Drain regions must be N-type. Therefore, two heavily doped N-type regions are created within the P-type substrate. These regions serve as the Source and Drain terminals of the transistor.

The Source and Drain are formed on opposite sides of the substrate using lithography and doping processes. This arrangement leaves a region of P-type material between them, where the conducting channel will later be formed.

Topic of Discussion	N-MOS (Enhancement-type NMOS transistor)
Channel Type	N-channel MOS — the conducting channel is formed by electrons (electron channel).
Source & Drain	Both are N-type regions embedded in the P-type substrate.
How They Are Formed	The N-type Source and Drain regions are created using the lithography process.

2.17 Channel Formation Between Source and Drain

After the Source and Drain regions are formed, the space between them remains P-type material. At this stage, there is no conducting path connecting the two N-type regions.

If a voltage is applied to the Drain, current cannot flow from the Source to the Drain because the electrons in the N-type Source and Drain are separated by the P-type region. The absence of a continuous N-type path prevents electron conduction between the two terminals.

Attempting to force current through this structure only strengthens the depletion region at the junctions, further blocking carrier movement. Therefore, a conducting channel must first be created.

To allow current flow, an N-type channel is formed in the P-type region between the Source and Drain. This channel acts as a bridge, providing a continuous path for electrons to move from the Source to the

Drain.

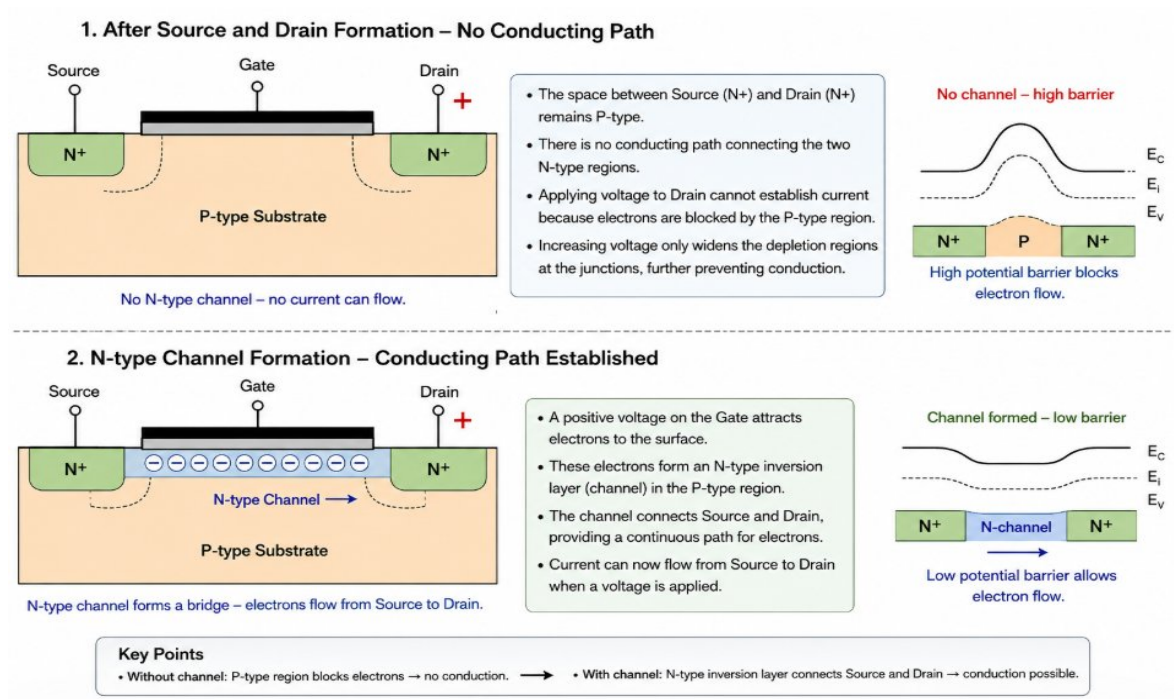


Figure 2.10: Channel Formation in NMOS – No Conducting Path vs. N-type Channel Established

- **Threshold Voltage** causes population inversion in the P-type region.
- V_{DS} makes current flow after population inversion has occurred.
- V_{GS} makes the MOSFET turn ON.

2.18 Gate Structure Above the Source–Drain Gap

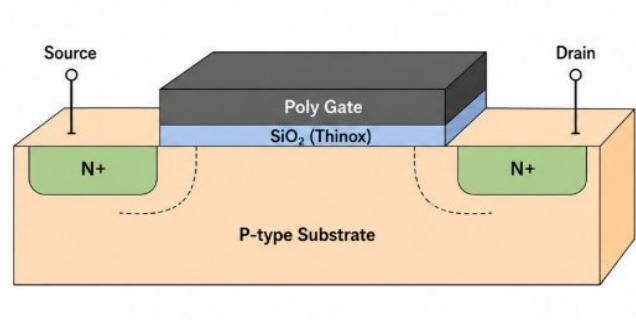
To create a conducting channel between the Source and Drain, a gate structure is built above the gap separating the two regions.

First, a very thin layer of silicon dioxide (SiO_2) is placed over the P-type region between the Source and Drain. This insulating layer is known as the thin oxide layer, or Thinox. It electrically separates the gate terminal from the semiconductor beneath it.

On top of the Thinox layer, the Gate electrode is formed. Although early MOSFETs used metal gates, modern devices typically use polycrystalline silicon (Poly) as the gate material.

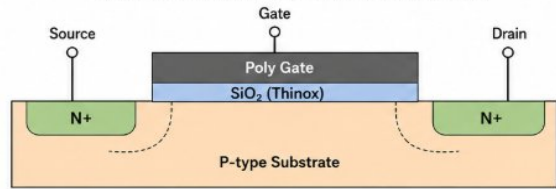
Together, the Thinox layer and the Poly gate form the gate stack, which controls the creation of the channel between the Source and Drain.

Gate Structure Built Above the Source–Drain Gap



- 1 A very thin layer of silicon dioxide (SiO_2) is grown over the P-type region between Source and Drain. This insulating layer is called the thin oxide layer, or **Thinox**.
- 2 On top of the Thinox layer, the Gate electrode is formed. Modern MOSFETs use polycrystalline silicon (**Poly**) as the gate material.
- 3 Together, the Thinox layer and the Poly gate form the **gate stack**, which controls the creation of the channel between Source and Drain.

Cross-Section View (Through the Channel Region)



Layer / Material	Purpose / Function
 Poly Gate	Gate electrode that controls the channel. (Modern MOSFETs use polycrystalline silicon.)
 SiO_2 (Thinox)	Very thin silicon dioxide layer. Electrically insulates the gate from the semiconductor.
 P-type Substrate	Semiconductor body where the channel is formed when a gate voltage is applied.

**Result:** This gate stack allows the gate voltage to electrically control the surface of the P-type region without any direct electrical contact, enabling the formation of an N-type channel between Source and Drain.

Figure 2.11: Gate Structure Built Above the Source–Drain Gap — 3D and Cross-Section View

2.19 Channel Formation by Gate Voltage — Population Inversion

The region beneath the gate oxide is initially P-type material and contains holes as the majority charge carriers. To understand channel formation, consider a simplified example where this region contains 100 holes.

When a positive voltage is applied to the gate, an electric field is created that attracts electrons toward the surface beneath the oxide layer. As the gate voltage increases, more electrons accumulate in this region.

If 100 electrons are attracted to the surface, they neutralize the 100 holes present there. At this point, the surface region becomes electrically neutral because the number of electrons and holes is equal.

When the gate voltage is increased further, additional electrons continue to accumulate. These extra electrons outnumber the holes, making electrons the majority carriers in that region. As a result, the surface of the P-type substrate begins to behave like N-type material.

This change, where the majority carrier population shifts from holes to electrons, is called population inversion. The inverted N-type region forms a conducting channel that connects the Source and Drain, allowing current to flow through the transistor.

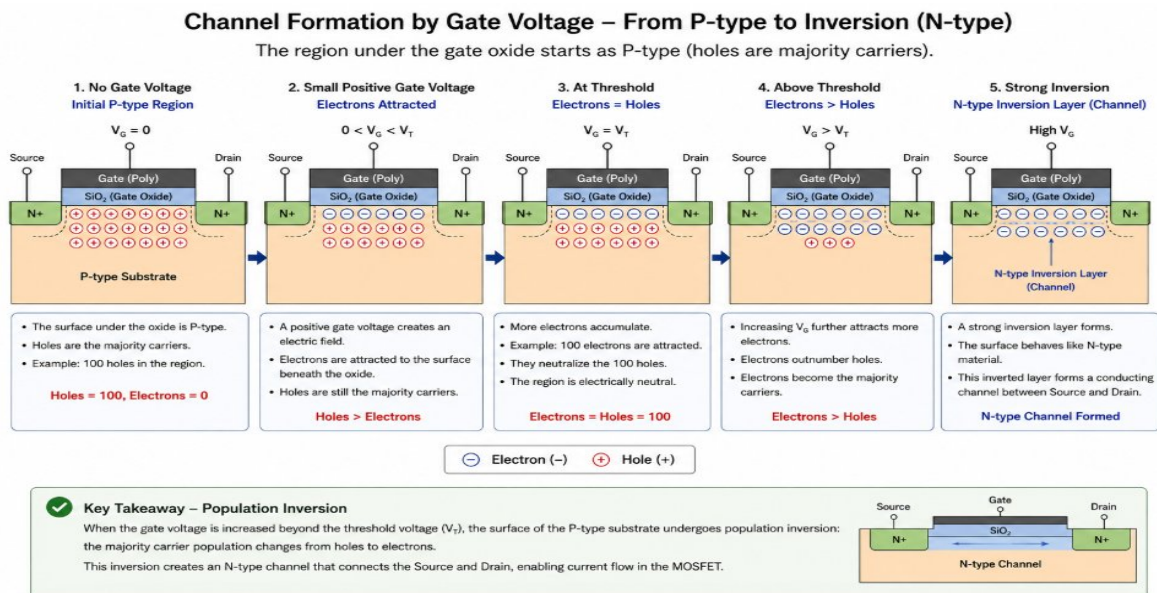


Figure 2.12: Channel Formation by Gate Voltage — From P-type to N-type Inversion

2.20 Threshold Voltage (V_{TH}) and Sub-Threshold Leakage

As the gate voltage increases, more electrons gather beneath the oxide and neutralize the holes in the P-type substrate. The minimum gate voltage required to form a strong inversion channel between Source and Drain is called the Threshold Voltage (V_{TH}).

When $V_{GS} \geq V_{TH}$, an N-type channel is formed and current can flow when V_{DS} is applied.

Ideal Textbook Model:

$V_{GS} < V_{TH}$	$I_D = 0$ (Device OFF)
$V_{GS} \geq V_{TH}$	$I_D > 0$ (Device ON)

However, real MOSFETs do not turn OFF abruptly. Even below V_{TH} , a very small current still flows due to the presence of a few carriers near the surface. This small current is called Sub-Threshold Leakage Current.

For simplified circuit analysis, this leakage current is usually ignored and the ideal ON/OFF model is assumed.

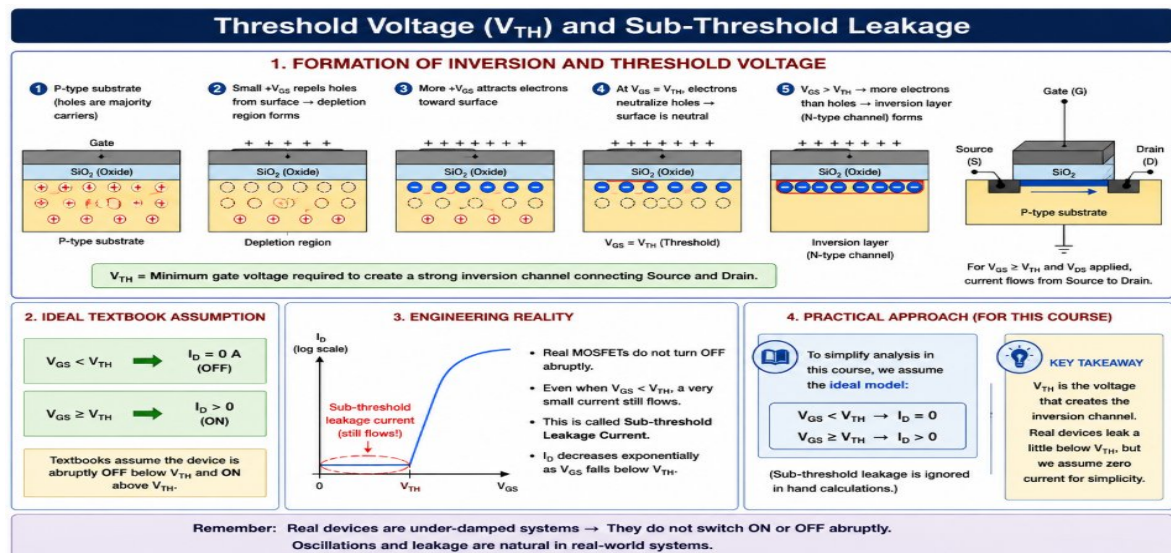


Figure 2.13: Threshold Voltage (V_{TH}) Formation and Sub-Threshold Leakage

2.21 MOSCAP and Real-World PC Voltages

The gate, oxide, and semiconductor form a MOS Capacitor (MOSCAP). The gate is the top conductor, the semiconductor is the bottom conductor, and the oxide is the dielectric between them. Its capacitance is:

$$C = \epsilon A / t_{ox}$$

Since capacitance is inversely proportional to oxide thickness, a thinner oxide gives higher capacitance. Higher capacitance means the gate field controls the channel more effectively, so less gate voltage is needed to reach inversion:

$$t_{ox} \downarrow \Rightarrow C \uparrow \Rightarrow V_{TH} \downarrow$$

A thick oxide can demand around 20 V to turn the transistor ON — far beyond what a modern PC provides (~ 1.1 V). So transistors must be designed with V_{TH} of only 0.2 V to 0.3 V, achieved by making the oxide layer extremely thin (typically ~ 20 nm in modern fabrication).

Thick oxide (old)	Low C, high V (~20 V) — cannot work in modern PCs.
Thin oxide (modern)	High C, low V (~0.2–0.3 V) — works at 1.1 V supply.

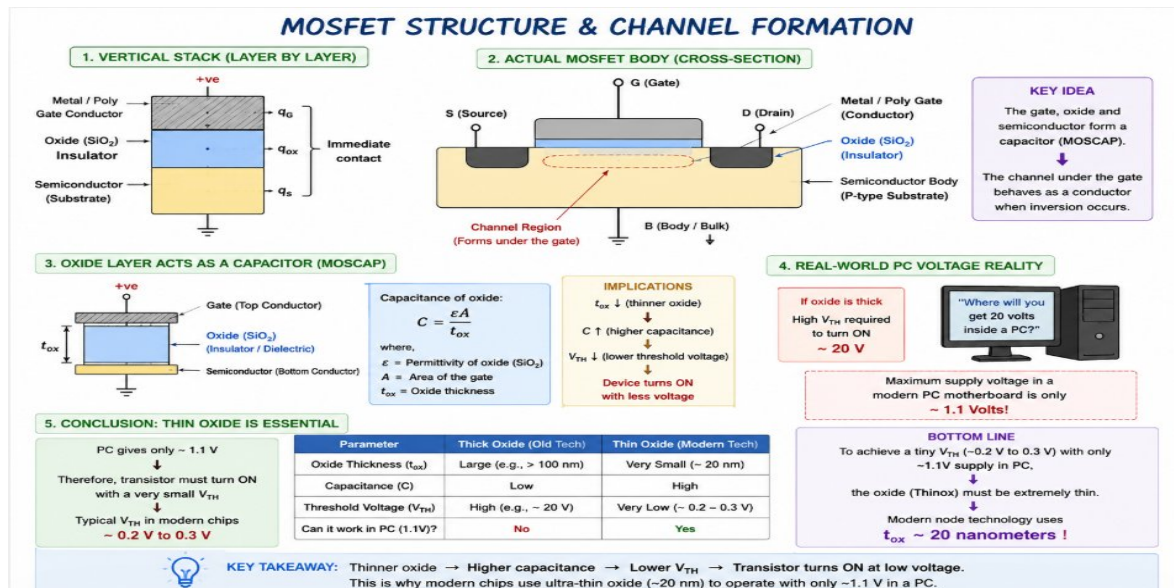


Figure 2.14: MOSFET Structure, MOSCAP Capacitance, and Real-World PC Voltage Requirements

2.22 Gate-to-Semiconductor Charge Regions in an NMOS

When a positive voltage is applied to the gate, charges appear in different parts of the MOS structure. The three relevant regions are the gate, the oxide interface, and the semiconductor beneath the oxide.

Gate Charge (Q_G)	The gate is a conductor. Applying a positive gate voltage places positive charge on the gate almost immediately.
Oxide/Interface Charge (Q_{SS})	A small amount of charge may become trapped at oxide surface states or interface states between the SiO_2 layer and the semiconductor. This is a small fixed/trapped charge associated with the oxide/interface region.
Inversion Channel Charge (Q_C)	Mobile electrons that accumulate at the surface beneath the oxide and form the conducting channel.
Depletion Charge (Q_D)	Immobile ionized acceptor charge left behind in the depletion region beneath the channel.

2.22.1 Charge-Balance Relation

Charge conservation requires that the positive charge above the oxide equals the total negative charge stored below it:

$$Q_G + Q_{SS} = Q_C + Q_D$$

Memory shortcut: Gate + oxide/interface charges on top = channel + depletion charges underneath.

- Once formed, the inversion channel itself behaves as a conductor.

Gate-to-Semiconductor Charge Regions in an NMOS

When a positive voltage is applied to the gate, charges appear in different parts of the MOS structure.

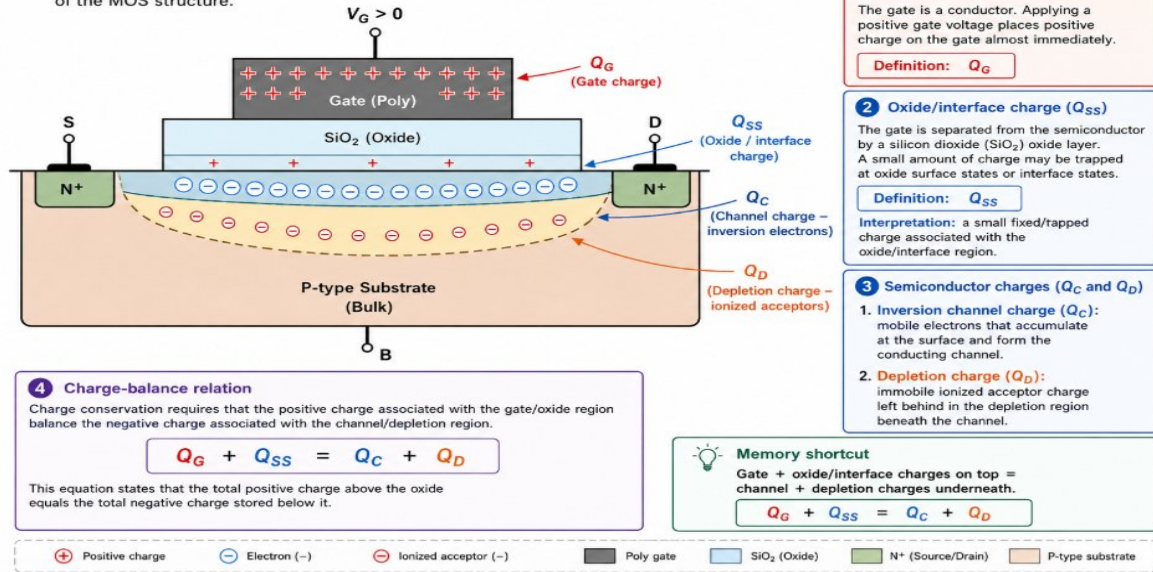


Figure 2.15: Gate-to-Semiconductor Charge Regions and Charge-Balance Relation in NMOS

2.23 Threshold Voltage Derivation from Charge Balance

Starting from the charge-balance relation $Q_G + Q_{SS} = Q_C + Q_D$, the threshold voltage can be derived by considering the condition at the point of inversion, where $Q_C = 0$:

$$V_t = Q_G / C_g = (Q_D - Q_{SS}) / C_g \text{ [if } Q_C = 0 \text{]}$$

For the general case where $Q_C \neq 0$:

$$V = Q_G / C_g = (Q_C + Q_D - Q_{SS}) / C_g$$

2.23.1 Practical Threshold Voltage (Equation 2.1)

In practice, V_t must also overcome built-in potential differences before the transistor reaches the edge of conduction. The full expression is:

$$V_t = Q_D / C_g + V_r - Q_{SS} / C_g + V_{dif}$$

Q_D / C_g	Depletion charge term – the only variable . Depends on doping concentration and applied voltages. All other terms depend only on construction and remain constant.
V_r	Voltage across the depletion region at the point of inversion (~0.2 V for P-type). Arises from the hole–electron gradient across the depletion region.
Q_{SS} / C_g	Oxide/interface trapped charge term. Reduces V_t slightly.
V_{dif}	Fermi level misalignment between the poly gate and the oxide – two different materials whose Fermi levels do not align, inducing a small extra voltage. Small in modern devices because silicon (poly) gates are used instead of metal.

2.23.2 Body Effect

Changing the substrate voltage alters Q_D , which is the only variable term in the threshold equation. This is the body effect. The modified threshold voltage accounting for body effect is:

$$V_t = V_{t0} + \gamma(V_{sb})^{\frac{1}{2}}$$

where γ is a constant that typically lies between 0.3 and 0.7 for MOS transistors (0.5 is commonly used). V_{t0} is the threshold voltage when $V_{sb} = 0$.

Why does V_{sb} change V_t ?

When V_{sb} is non-zero, the depletion region under the gate widens. A wider depletion region exposes more ionized acceptor charge, increasing Q_D . Since Q_D is the only variable term in the threshold equation, a larger Q_D requires a larger gate voltage to reach inversion — so V_t increases. In short: V_{sb} changes Q_D , which changes V_t .

Why is $Q_C = 0$ at V_t ?

V_t is defined as the minimum gate voltage needed to just reach strong inversion. At this exact boundary point, the channel is just being formed and has not yet accumulated any significant mobile charge. The entire gate voltage at this moment is used to push holes away (building Q_D) and overcome interface charge (Q_{ss}) — not to fill the channel with electrons yet. Once V_{GS} exceeds V_t , Q_C starts building up.

Adjusting V_t Through Doping Concentration

The threshold voltage can be tuned during fabrication by changing the doping concentration of the P-type substrate:

Heavy P-type doping (P)	More holes are present in the substrate. More electrons are needed to neutralize them and reach inversion — higher V_t .
Light P-type doping (P)	Fewer holes are present. Fewer electrons are needed to reach inversion — lower V_t .

This gives chip designers a direct handle on V_t — by selecting the doping level during fabrication, the transistor can be made to turn ON at a higher or lower gate voltage as required.

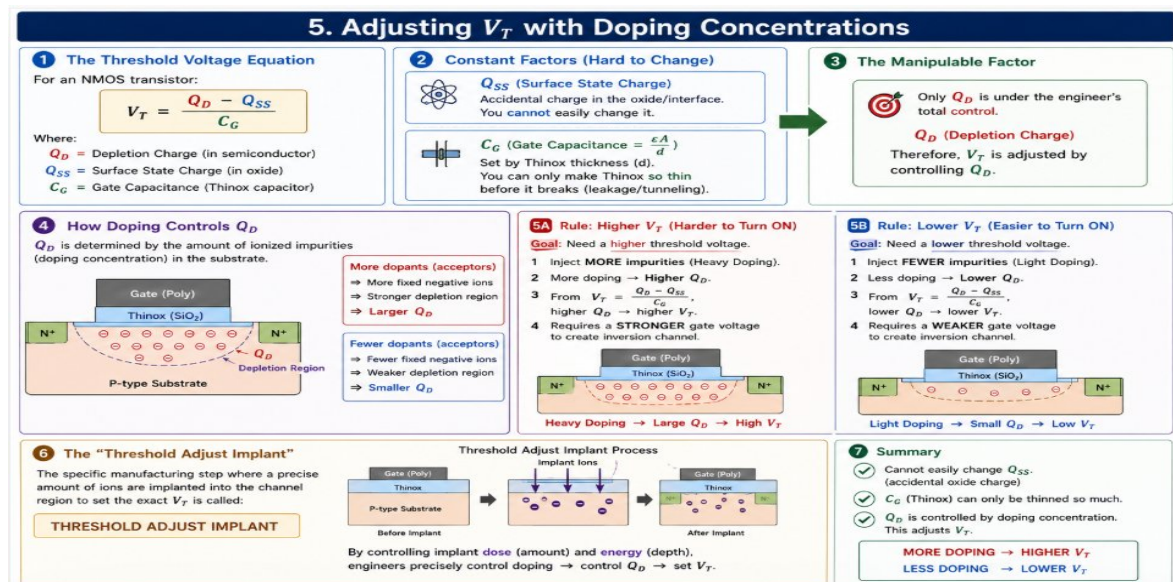


Figure 2.18: Adjusting V_T with Doping Concentration — Q_D Control and Threshold Adjust Implant

2.23.4 Multi-Threshold CMOS and Depletion-Type NMOS

Since doping concentration directly controls V_t , chip designers deliberately use different doping levels across a chip to create transistors with different threshold voltages. This technique is called **Multi-Threshold CMOS (MTCMOS)**.

Enhancement-type NMOS	Standard device. No channel at $V_{GS} = 0$. Requires a positive V_t to turn ON. Most common in logic circuits.
Depletion-type NMOS	An n-type implant is placed directly between Source and Drain during fabrication. A channel already exists at $V_{GS} = 0$. The device is ON by default and requires a negative gate voltage to turn OFF. V_t is negative.

- The boundary between enhancement and depletion behavior is called the **demarcation line** — the point where $V_t = 0$.

2.23.5 Body Connections and Body Effect Recap

NMOS body	Connected to Ground (0 V) to keep the source–body junction reverse-biased.
PMOS body	Connected to V_{DD} (highest supply) for the same reason.
Standard V_{sb}	Source–substrate voltage is 0 V in standard connections for both NMOS and PMOS.
When $V_{sb} \neq 0$	Depletion region widens Q_D increases V_t increases. This is the body effect.

Body effect formula:

$$V_t = V_{t0} + \gamma(V_{sb})^{\frac{1}{2}}$$

- **Recap:** V_t Q_D – threshold voltage is dynamically adjustable through doping concentration, oxide thickness, and body voltage.

2.24 Gate Capacitance (C_G) and Thinox

The gate structure behaves like a parallel-plate capacitor with capacitance:

$$C_G = \epsilon A / d$$

where ϵ is the oxide permittivity, A is the gate area, and d is the oxide thickness. For modern low-voltage circuits, V_T must be kept small, which requires a large C_G . A large C_G is achieved by making d extremely small.

The ultra-thin SiO layer grown directly beneath the gate is called **Thinox**. It provides the large gate capacitance needed for effective control of the semiconductor surface.

Thinox (thin oxide)	Large C_G strong gate control low V_T . Channel can form here.
Field oxide (thick oxide)	Acts as a plain insulator. C_G is too small for channel formation. No channel can form here.

- Golden Rule: the conducting channel can form **only** beneath a region covered by Thinox.

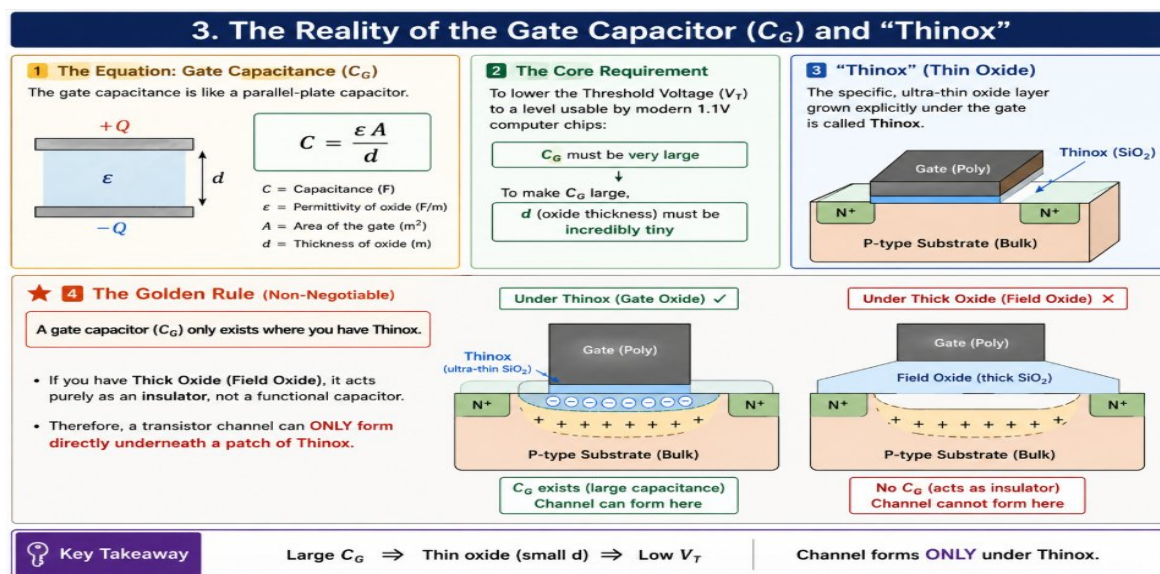


Figure 2.16: Gate Capacitance, Thinox vs. Field Oxide – Where the Channel Can Form

2.25 Layout Reality: Where a Transistor Actually Forms

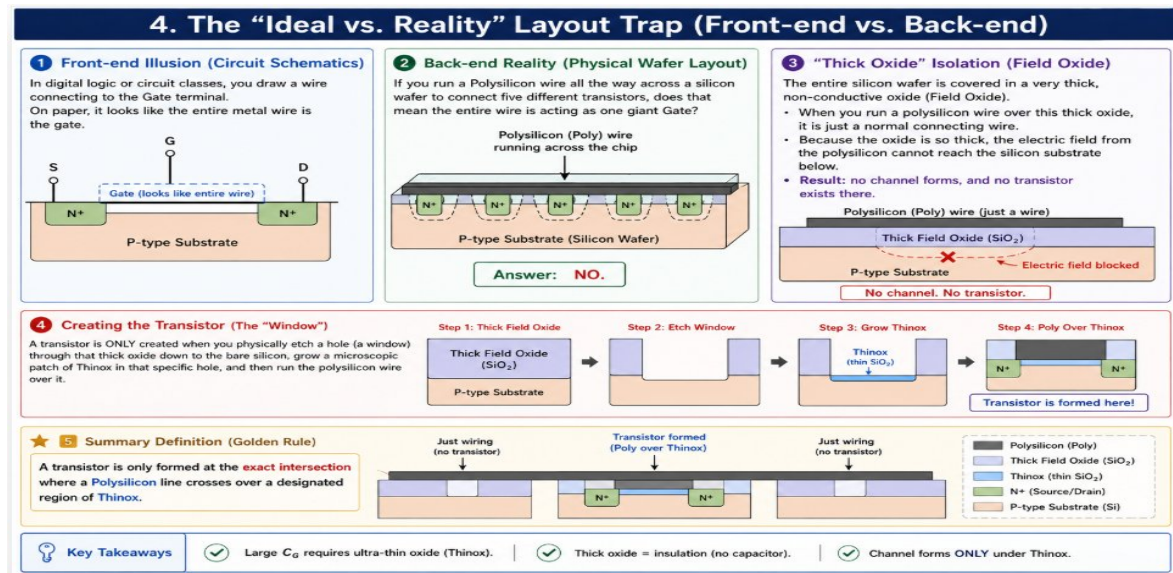
A common misconception in VLSI layout is that an entire polysilicon wire acts as a gate. In a schematic it looks that way, but on the physical wafer the reality is different.

The entire silicon wafer surface is covered by a thick Field Oxide. When a polysilicon wire runs over this thick oxide, it is simply a wire. The oxide is too thick for the electric field to reach the substrate below, so no channel forms and no transistor exists there.

A transistor is created only by etching a window through the field oxide down to bare silicon, growing a microscopic patch of Thinox in that window, and then running the polysilicon wire over it.

Poly over Field Oxide	Just a wire. No channel. No transistor.
Poly over Thinox	Gate of a transistor. Channel forms beneath. Transistor exists here.

- A transistor forms **only** at the exact intersection where a polysilicon line crosses a designated Thinox region. Everything else is wiring.



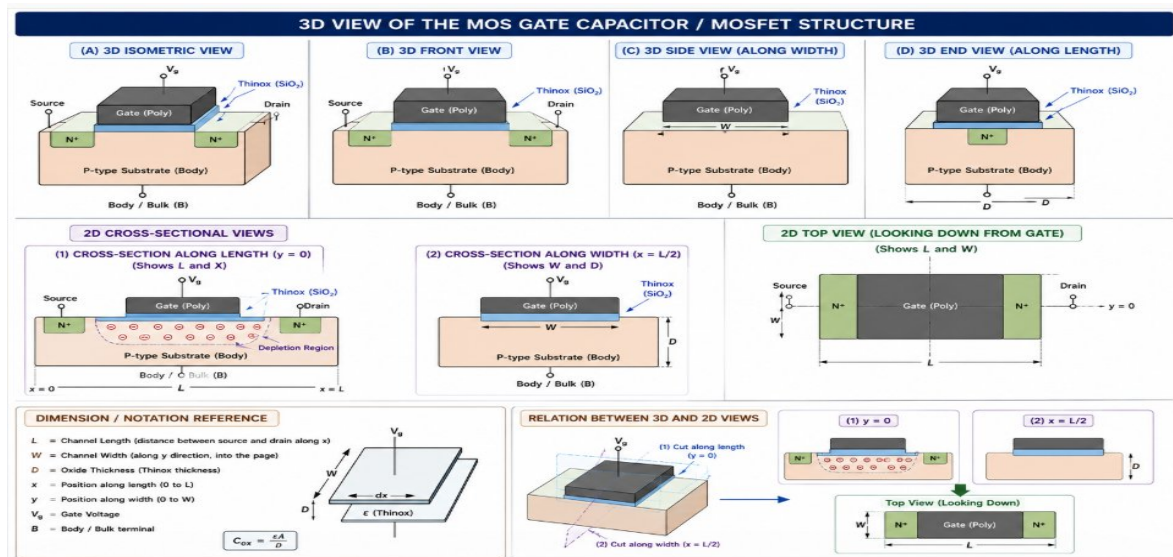


Figure 2.19: 3D View of MOS Gate Capacitor/MOSFET Structure — Isometric, Front, Side, End, and 2D Cross-Sections

2.27 I_{DS} Derivation in the Resistive Region

When $V_{GS} \geq V_t$ and $V_{DS} < V_{GS} - V_t$, a continuous inversion channel exists from Source to Drain. The channel voltage increases linearly from 0 V at the Source to V_{DS} at the Drain.

2.27.1 Step 1 – Elemental Capacitor

Consider a thin slice of the channel of length dx at position x from the Source. This slice behaves like a small parallel-plate capacitor. Its capacitance is:

$$C = \epsilon W dx / D$$

where ϵ is the permittivity of the oxide, W is the channel width, and D is the oxide thickness.

2.27.2 Step 2 – Voltage Across the Elemental Capacitor

The voltage at the drain end is $V_{gd} = V_{GS} - V_{DS}$. The voltage in excess of V_t controlling inversion at position x is:

$$v = V_{gd} + (x/L) V_{DS} - V_t = V_{GS} - V_{DS} + (x/L) V_{DS} - V_t$$

2.27.3 Step 3 – Charge on the Elemental Capacitor

The charge $q = C \cdot v$ on this slice is:

$$q = (\epsilon W dx / D)(V_{GS} - V_{DS} + (x/L) V_{DS} - V_t)$$

2.27.4 Step 4 – Total Charge Q in the Entire Channel

Integrate q over the full channel from $x = 0$ to $x = L$:

$$Q = \int_0^L (\epsilon W/D)(V_{GS} - V_{DS} + (x/L) V_{DS} - V_t) dx$$

Split into constant and variable parts:

$$Q = (\epsilon W/D) \int_0^L (V_{GS} - V_t) dx + (\epsilon W/D) \int_0^L ((x/L) V_{DS} - V_{DS}) dx$$

First integral – $(V_{GS} - V_t)$ is constant w.r.t. x :

$$\int_0^L (V_{GS} - V_t) dx = (V_{GS} - V_t) \cdot L$$

Second integral – factor out V_{DS} , integrate $(x/L - 1)$:

$$\int_0^L (x/L - 1) V_{DS} dx = V_{DS} [x^2/(2L) - x]_0^L = V_{DS}(L/2 - L) = -V_{DS} L/2$$

Combining and factoring L/D :

$$Q = (\epsilon WL/D) [(V_{GS} - V_t) - V_{DS}/2]$$

2.27.5 Step 5 – Transit Time of an Electron

The electric field across the channel is V_{DS}/L , so electron velocity $= \mu_n V_{DS}/L$. The transit time t is:

$$t = L / (\mu_n V_{DS}/L) = L^2 / (\mu_n V_{DS})$$

μ_n is the electron mobility – velocity per unit electric field ($m^2/V \cdot s$).

2.27.6 Step 6 – Final I_{DS} Equation

Substituting Q and t into $I_{DS} = Q/t$:

$$I_{DS} = (\epsilon W L / D) [(V_{GS} - V_t) - V_{DS} / 2] \div L^2 / (\mu_n V_{DS})$$

$$I_{DS} = (\epsilon W \mu_n / LD) [(V_{GS} - V_t) V_{DS} - V_{DS}^2 / 2]$$

This is Equation 2.2 – the drain current equation valid in the resistive (linear/triode) region.

2.27.7 Simplified Form for Small V_{DS}

When $V_{DS} \ll V_{GS} - V_t$, the $V_{DS}^2/2$ term is negligible:

$$I_{DS} \approx (\epsilon W \mu_n / LD) (V_{GS} - V_t) V_{DS}$$

This gives a linear (ohmic) relationship between I_{DS} and V_{DS} for constant V_{GS} .

2.28 Transconductance and Dependence on Permittivity

From the I_{DS} equation, current depends on both V_{GS} and V_{DS} . In the linear region, $I_{DS} \propto V_{GS} \cdot V_{DS}$. As V_{DS} approaches $V_{GS} - V_t$, behavior transitions toward $I_{DS} \propto V^2$ at saturation onset.

2.28.1 Conductance

Conductance g is defined as:

$$g = I / V$$

In the MOSFET channel, g depends on the permittivity of the oxide, which determines how effectively the gate field controls carriers below.

2.28.2 Transconductance

Transconductance g_m measures how effectively V_{GS} controls I_{DS} :

$$g_m = \partial I_{DS} / \partial V_{GS}$$

Differentiating the simplified I_{DS} with respect to V_{GS} :

$$g_m = \epsilon W \mu_n V_{DS} / (LD)$$

(Permittivity)	Permittivity of the oxide. Higher ϵ stronger gate-to-channel coupling higher g_m . This is why high- ϵ dielectrics are used in modern transistors.
W (Width)	Wider channel more parallel current paths higher g_m .
μ_n (Mobility)	Electron mobility. Higher mobility faster electrons higher g_m . Depends on material and temperature.
L (Length)	Shorter channel shorter transit time higher g_m . Key reason modern chips use nm-scale channel lengths.
D (Oxide Thickness)	Thinner oxide higher E/D coupling higher g_m . Balanced against leakage/tunneling risk.

• $g_m = W_n/LD$ is the device's fundamental gain parameter. It depends directly on permittivity — this is why high- ϵ dielectrics boost g_m without requiring dangerously thin oxide.

Appendix: Job Written & Viva Questions

Questions likely to appear in written tests and viva boards based on the topics covered in this class.

A.1 NMOS Structure & Channel Formation

Q: Why is enhancement-type NMOS used as the standard device for analysis?

A: Its majority carriers are electrons — real physical particles that are straightforward to visualize. PMOS uses holes (absence of electrons), making current directions counterintuitive. Depletion-type devices already have a channel at $V = 0$, reversing the usual analysis conditions.

Q: Why can current not flow between Source and Drain immediately after they are formed?

A: The space between the two N regions remains P-type material. There is no continuous N-type path, so electrons in the Source and Drain are blocked by the P-type barrier. Applying V only widens the depletion regions further, increasing the barrier.

Q: What is the role of the gate structure in an NMOS transistor?

A: The gate structure — consisting of the Thinox (SiO_2) layer and the polysilicon gate — sits above the P-type region between Source and Drain. Applying a positive V creates an electric field that attracts electrons to the surface, eventually forming an N-type inversion channel that bridges Source and Drain.

Q: What is population inversion and when does it occur?

A: Population inversion occurs when the number of electrons at the P-type substrate surface exceeds the number of holes. As V increases, electrons accumulate and first neutralize the holes (surface becomes intrinsic), then outnumber them. The surface then behaves as N-type material — the inversion channel is formed.

A.2 Threshold Voltage & Charge Balance

Q: Derive the basic expression for threshold voltage V_t from the charge balance equation.

A: Starting from $Q_p + Q_n = Q_p + Q_n$. At the point of inversion, $Q_p = 0$, so $Q_n = Q_p - Q_p$. Dividing by C gives $V = Q_p/C = (Q_p - Q_n)/C$. In practice, additional terms for built-in potentials give: $V = Q_p/C + V_{bi} - Q_n/C + V_{bi}$.

Q: Why is $Q_n = 0$ at the threshold voltage?

A: V is defined as the minimum gate voltage to just reach strong inversion. At this exact boundary, the channel is only just forming and has not yet accumulated significant mobile charge. The entire gate voltage is consumed pushing holes away (building Q_p) and overcoming Q_n . Mobile channel charge Q_n starts accumulating only once V exceeds V_t .

Q: Which term in the threshold voltage equation is the only variable, and why?

A: Q (depletion charge) is the only variable term. It depends on the doping concentration of the substrate and the applied body voltage. All other terms — V , Q/C , and V — are determined by material properties and construction and remain constant for a given device.

Q: What is V and why is it small in modern MOSFETs?

A: V is the Fermi level misalignment voltage arising because the polysilicon gate and the oxide are different materials with different Fermi levels. This mismatch induces a small built-in voltage. It is small in modern devices because polysilicon (silicon-based) gates are used instead of metal, reducing the Fermi level difference.

Q: What is the body effect and how does it affect V_t ?

A: The body effect is the change in V caused by a non-zero source–substrate voltage V . Increasing V widens the depletion region under the gate, increasing Q . Since $V \propto Q$, a larger Q requires more gate voltage to reach inversion, so V rises. Expressed as: $V = V + (V)^{1/2}$.

A.3 Gate Capacitance, Thinox & Layout

Q: Why must the gate oxide (Thinox) be extremely thin in modern MOSFETs?

A: Gate capacitance $C = A/D$. To achieve a large C — which is needed to lower V to 0.2–0.3 V for 1.1 V operation — the oxide thickness D must be minimised. Modern fabrication uses oxide thicknesses on the order of ~20 nm.

Q: What is the 'Golden Rule' of MOSFET layout regarding Thinox?

A: A transistor channel forms only at the exact intersection where a polysilicon line crosses a designated Thinox region. Polysilicon running over thick field oxide is simply a wire — the oxide is too thick for the gate field to reach the substrate, so no channel forms there.

Q: What is Multi-Threshold CMOS (MTCMOS) and how is it achieved?

A: MTCMOS is a design technique where transistors with different threshold voltages are used on the same chip. Since $V \propto Q$ and Q is controlled by doping concentration, different doping levels are implanted in different regions during fabrication (Threshold Adjust Implant), creating high- V (low leakage) and low- V (high speed) devices on the same die.

Q: What is the difference between a depletion-type and enhancement-type NMOS?

A: Enhancement-type NMOS has no channel at $V = 0$ and requires a positive V to turn ON. Depletion-type NMOS has an N-type implant between Source and Drain during fabrication, so a channel already exists at $V = 0$. It requires a negative gate voltage to turn OFF. V is negative for depletion-type devices.

A.4 I_{DS} Derivation & Transconductance

Q: State the drain current equation (Eq. 2.2) for the resistive region and explain each term.

A: $I = (W/LD)[(V - V_t)V - V^2/2]$. Here ϵ is oxide permittivity, W is channel width, μ_n is electron mobility, L is channel length, D is oxide thickness, V_g is gate-to-source voltage, V_t is threshold voltage, and V_{ds} is drain-to-source voltage.

Q: How is the elemental capacitor concept used to derive I_{DS} ?

A: The channel is modelled as an infinite series of thin capacitor slices of length dx . Each slice has capacitance $C = W dx/D$ and a local voltage v in excess of V_t . The charge on each slice is $q = Cv$. Integrating q from $x=0$ to $x=L$ gives the total channel charge Q . Using $Q = I \cdot t$ and the transit time $t = L^2/(V_{ds} \mu_n)$ yields the final equation.

Q: What simplification is valid for very small V_{DS} , and what does it imply?

A: When $V_{ds} \ll V - V_t$, the $V^2/2$ term is negligible, giving $I \approx (W/LD)(V - V_t)V_{ds}$. This is linear in V_{ds} , confirming that the MOSFET behaves as a voltage-controlled resistor (ohmic behavior) in this region.

Q: Define transconductance g_m and write its expression.

A: Transconductance $g = \partial I / \partial V_g$, the rate of change of drain current with gate voltage. From the simplified I equation: $g = W \mu_n V_{ds} / (LD)$. It is the fundamental gain parameter of the transistor.

Q: How does permittivity affect transconductance, and what is the practical implication?

A: $g = W \mu_n V_{ds} / (LD)$ is directly proportional to ϵ . Higher permittivity means stronger capacitive coupling between gate and channel, giving better current control per volt of gate voltage. This is why modern transistors use high- ϵ (high permittivity) dielectric materials as the gate oxide instead of pure SiO_2 — to boost g without requiring dangerously thin oxide layers.

A.5 Viva One-Liners (Quick Fire)

What does MOSCAP stand for?	MOS Capacitor — the gate, oxide, and semiconductor acting as a parallel-plate capacitor.
What is Thinox?	The ultra-thin SiO_2 layer grown specifically beneath the gate to provide large C .
What is the charge-balance equation?	$Q_g + Q_{ox} = Q_{dep} + Q_{it}$
What is Q_{dep}?	Depletion charge — immobile ionized acceptor ions in the depletion region. The only variable term in V_t .
What is Q_{it}?	Oxide/interface trapped charge — small fixed charge at the SiO_2 -semiconductor interface.
Why is V_t dynamically adjustable?	Because V_t , Q_{dep} , and Q_{it} can be changed by doping concentration or body voltage.

What happens when poly runs over field oxide?	It is just a wire. No C, no channel, no transistor.
What is the body effect formula?	$V = V + (V)^{1/2}$
What is electron mobility ?	Electron velocity per unit electric field ($\text{m}^2/\text{V}\cdot\text{s}$).
What is transconductance g?	$W V / (LD)$ – measures gate control over drain current.
How do you increase g?	Increase W , decrease L or D , use high-mobility material.
What is the demarcation line?	The boundary where $V = 0$, separating enhancement-type ($V > 0$) from depletion-type ($V < 0$) devices.
Why use high- dielectrics?	To increase C (and hence g) without making oxide thickness D dangerously small.